

CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 8

NMAP LIVE HOST DISCOVERY

Aim

To discover live or active hosts in a network using Nmap scans such as ARP, ICMP, TCP and UDP.

Objective

To understand how Nmap identifies active systems before performing further network scanning.

Software Required

- Nmap
- TryHackMe Nmap room / AttackBox
- Computer with network access

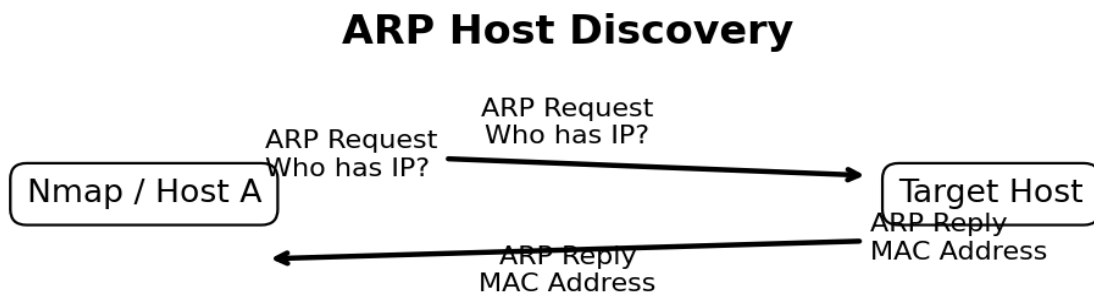
Introduction

Nmap (Network Mapper) is a free and open-source network scanning tool used to discover live hosts and identify network services. The lab manual introduces ARP, ICMP, TCP and UDP as host-discovery approaches.

Nmap host discovery aims to identify active systems before more extensive scanning. The official Nmap documentation describes ARP for local Ethernet targets and ICMP, TCP and UDP probes as host-discovery techniques.

Source: Nmap official documentation.

ARP Host Discovery



A response indicates the host is reachable on the local Ethernet network.

Figure 1. Simplified ARP discovery diagram. Source concept: Nmap host-discovery documentation.

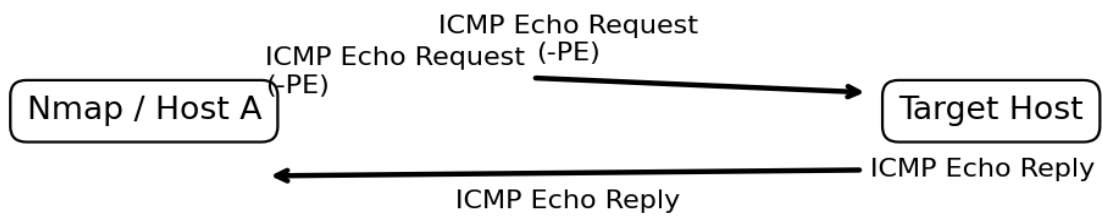
ARP uses an ARP request to determine the hardware/MAC address associated with a target IP on the local Ethernet network. A response indicates the host is reachable on that local network.

Command: `nmap -PR -sn TARGETS`

The lab manual notes that ARP discovery works on the same subnet because ARP is a link-layer protocol.

ICMP Host Discovery

ICMP Echo Discovery



Echo reply indicates that the target host is responding.

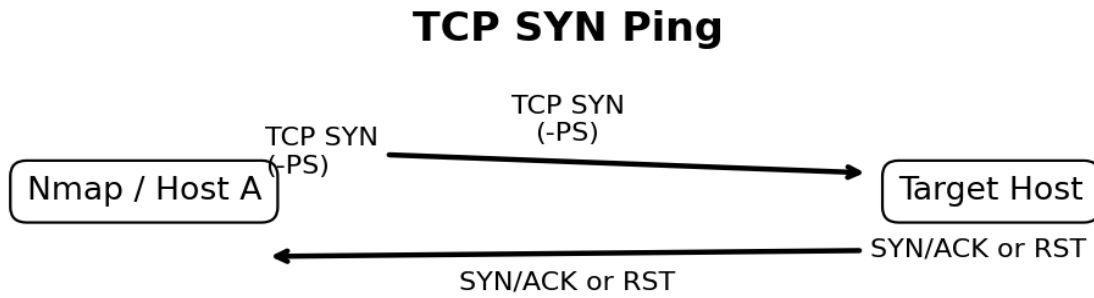
Figure 2. Simplified ICMP Echo discovery diagram.

Nmap can send an ICMP Echo Request (Type 8) and look for an ICMP Echo Reply (Type 0). The reply indicates that the target is responding.

Command: `nmap -PE -sn TARGETS`

The lab manual also identifies -PP for ICMP Timestamp and -PM for ICMP Address Mask discovery.

TCP SYN Host Discovery



A TCP response indicates that the host is available.

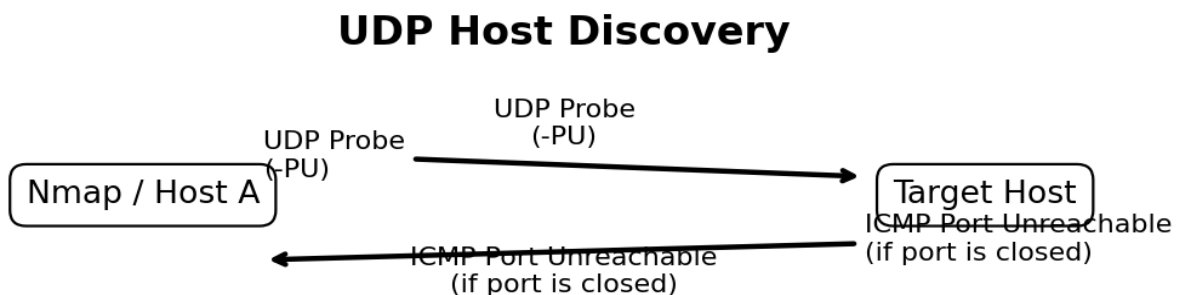
Figure 3. Simplified TCP SYN discovery diagram.

TCP SYN Ping sends a TCP packet with the SYN flag set to a selected port. A response such as SYN/ACK or RST can demonstrate that the host is responding.

Command: `nmap -PS23 TARGETS`

The lab manual identifies TCP SYN Ping as not requiring a privileged account and TCP ACK Ping as requiring a privileged account in the described context.

UDP Host Discovery



An ICMP port-unreachable response indicates the host is up.

Figure 4. Simplified UDP discovery diagram.

UDP Ping sends a UDP probe to a selected port. If the port is closed, an ICMP Port Unreachable response can indicate that the host is active.

Command: `nmap -PU40125 TARGETS`

Important Nmap Commands

Purpose	Option
Host discovery only	<code>nmap -sn TARGETS</code>
ARP discovery	<code>nmap -PR -sn TARGETS</code>
ICMP Echo	<code>nmap -PE -sn TARGETS</code>
ICMP Timestamp	<code>nmap -PP -sn TARGETS</code>
ICMP Address Mask	<code>nmap -PM -sn TARGETS</code>

Purpose	Option
TCP SYN Ping	nmap -PS23 TARGETS
TCP ACK Ping	nmap -PA80 TARGETS
UDP Ping	nmap -PU40125 TARGETS
Disable reverse DNS	nmap -n TARGETS
Reverse DNS lookup	nmap -R TARGETS

Observation

1. ARP can discover hosts within the same subnet.
2. ICMP Echo can be used to check whether a host responds.
3. TCP SYN Ping checks a host by sending a SYN probe to a TCP port.
4. TCP ACK Ping uses an ACK probe and, in the lab context, requires a privileged account.
5. UDP Ping can identify an active host through an ICMP Port Unreachable response from a closed UDP port.
6. The -sn option performs host discovery without proceeding to a port scan.

Observation Table

Scan Type	Option	Expected observation
ARP	-PR	ARP reply / MAC information from local host
ICMP Echo	-PE	ICMP Echo Reply
ICMP Timestamp	-PP	ICMP Timestamp Reply
ICMP Address Mask	-PM	ICMP Address Mask Reply
TCP SYN	-PS	SYN/ACK or RST response
TCP ACK	-PA	RST response can indicate a live host
UDP	-PU	ICMP Port Unreachable can indicate a live host

Result

Thus, live host discovery using Nmap was successfully studied using ARP, ICMP, TCP and UDP techniques.

Conclusion

Nmap provides multiple techniques for discovering active hosts. ARP is mainly useful on the local Ethernet network, while ICMP, TCP and UDP probes provide additional methods for determining whether a target host is responding.

Diagram Note

The four diagrams in this PDF are original educational diagrams created for this lab record from the protocol behavior described in the uploaded lab manual and the official Nmap documentation. They are not copied screenshots from another website.

Web source: Nmap Network Scanning – Host Discovery.